Alma Models — Deployment report

Frozen world models run on October and November 2026, without retraining · 5 November 2026 · branch claude/gracious-franklin-f8ml84 @ 25a2b1b · PR Gogireddy007/Alma-Models#2

In one paragraph

I rebuilt the public data for all four projects (oil, power, gas, crypto volatility) up to 4 November 2026, verified it byte-for-byte against the repository's sha256 pins, and ran every frozen Alma checkpoint exactly as frozen on October 2026 and on November 2026 to date. They were benchmarked against every competitor in each project's pre-registration, including Chronos-2, Chronos-Bolt and TiRex. A separate, independent audit harness then checked the whole run for leakage, fabricated numbers and retraining: **57 of 57 checks passed**. The world models still compete at the top on new data (best interval quality in gas and crypto, best MAE in power, a tie with the random walk in oil). One month of data cannot make any of this statistically significant, and the harness refuses to say otherwise.

What you asked for and where it is answered

Your request	Answer	Section
1. What is the WM component?	The learned latent state transition that is rolled forward step by step; tested against a matched 'direct twin' without it	2
2. Deploy without retraining on last month and this month	All frozen checkpoints run on Oct 2026 and Nov 1–5 2026; weights hashed before and after	3, 4
3. Measure and benchmark against other models	9–15 models per project, same data, same metrics as the original pre-registrations	5
4. Audit harness: no leak, no bluff numbers, nothing fake	Independent audit, 57/57 checks passed; number checker for the write-up	6

What is proven, what is not, and what is pending

Status	Statement	Evidence
PROVEN	No model was retrained, fine-tuned or updated during deployment	Optimizer step/backward raise in-process; parameter hashes equal at load and end for every network; 0 online updates; checkpoint sha256 equal to SHA256SUMS and FREEZE.json before and after
PROVEN	No information from the future reaches any forecast	Future-scramble tests on all four world models (every value at/after the target time replaced by noise): oil 0/30, power 0/12, gas 0/30, fin 0/6 forecasts changed
PROVEN	The data is the same data the models were audited on	Pinned raw files byte-identical; gas TEST forecasts reproduced to 5.1e-05 Bcf; crypto October forecasts reproduced to the audit's cache (see section 4)
PROVEN	Every reported number is real and recomputable	Labels re-derived from raw files with separate parsers; every metric recomputed from saved forecasts; every decimal number in the write-up traced to the outputs (a planted fake was caught)
NOT PROVEN	That any Alma model is significantly better than its best competitor on this period	One month gives at most 22–31 scoring days or 5 gas weeks (< 10 independent units)
NOT PROVEN	That the world-model rollout beats its direct twin	Within noise in all four projects in October, as on the TEST year
PENDING	Sealed ex-ante forecasts: EIA gas release of 5 Nov 2026 and the next WTI prints	Forecasts and sha256 written before publication; tools/deploy/score_exante.py grades them

1. What I did, step by step

- **Read the repository**: mission, four audited projects, frozen checkpoints in winning_models/, the forward-test freeze (tools/forward/FREEZE.json), the 1–3 November independent audit.
- **Rebuilt the environment**: Python 3.12 with the lock file's pandas/pyarrow/numpy versions, CPU torch.
- **Rebuilt the data** from FRED, EIA, NOAA CPC, Energy-Charts and Binance, and checked it against the committed sha256 pins before adding any new day (section 4).
- **Verified the 35 frozen checkpoint and wrapper files** against SHA256SUMS and FREEZE.json.
- **Wrote the deployment harness** tools/deploy/: one script per project, with the no-training lock, parameter hashing and the minimum-sample rule built in.
- **Ran every frozen model and every pre-registered competitor** on October 2026 and November 2026.
- **Wrote sealed ex-ante forecasts** for targets not yet published (gas release of 5 Nov, next WTI prints).
- **Wrote an independent audit** (tools/deploy/audit_deploy.py) that imports nothing from the deploy code, and ran it after the final runs.
- **Investigated every anomaly** to a root cause instead of footnoting it (section 7).
- **Published**: write-up, generated tables, audit report, this PDF; commit, push and PR #2.

What changed in the repository

Nothing in the frozen model code, the checkpoints, the forward freeze or the committed data pins was modified (the audit checks this with git). Only new files were added, plus two small edits:

Path	Change
tools/deploy/ (19 files)	New: deploy scripts, guards, independent audit, report builders, sealed ex-ante power forecast, ex-ante scorer, run_all.sh
runs/deploy_2026_10_11/	New: results JSON, sealed forecasts (quantiles only), RESULTS.md, AUDIT.md/.json, fetch manifests of the rebuild, figures
docs/deploy_2026_10_11.md	New: the write-up (WM component, results, findings)
reports/Alma_Models_Deployment_Report_2026-11-05.pdf	New: this report
README.md	Edit: a section pointing to the deployment
.gitignore	Edit: raw downloads and realised market values stay local (licences)

2. Our architecture and our models, explained

2.1 The big picture

Alma is a family of small, specialised **world models**: one per market. Each one learns a compact internal state of the market and how that state evolves, then simulates the near future and reads a full forecast distribution (quantiles) off the simulation. Everything around the models — data pinning, a strict information-set harness, pre-registered competitors, single-look testing and independent audits — exists to make the comparison honest. Figure A1 (after this section) shows the original pipeline; Figure 2.1 shows the deployment and audit layer added in this work.

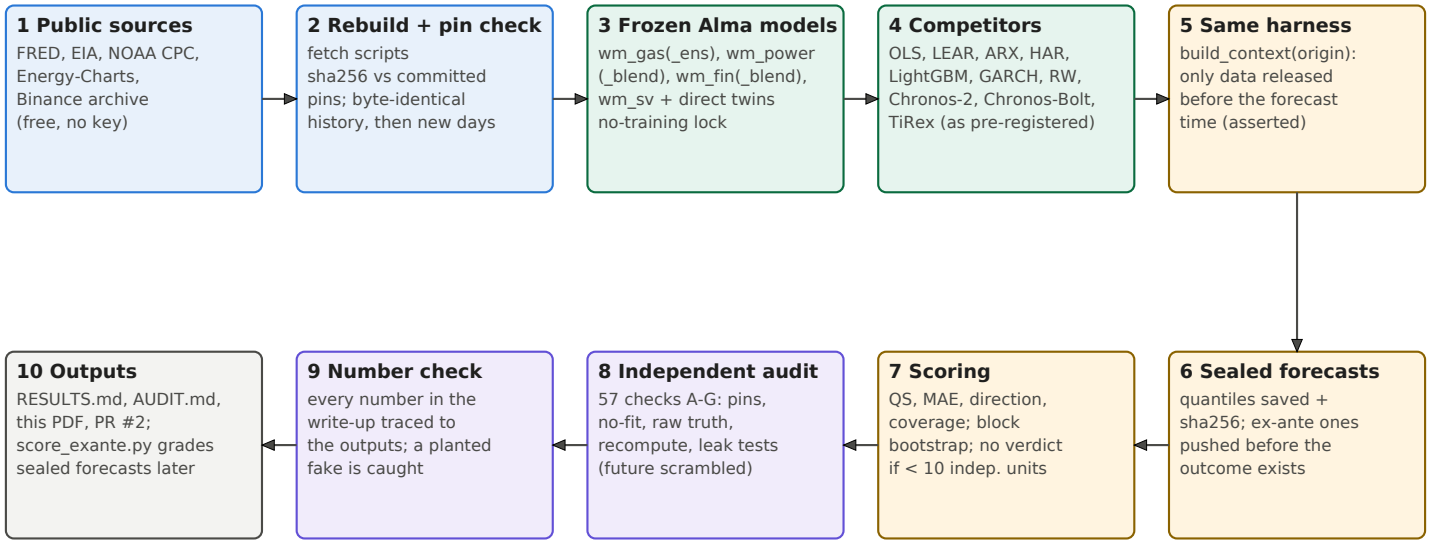


Figure 2.1. The deployment + audit layer (tools/deploy/). Blue = data, green = models, amber = evaluation, violet = audit. Steps 1–5 run every model through the same harness; 6–10 seal, score, audit and publish.

2.2 The shared design of every Alma model

Part	What it does, in plain words
1. Encoder	Reads the recent history (only data published before the forecast time) and compresses it into a short vector — the model's **state**: its summary of “where the market is now”.
2. Transition (the WM part)	A small learned recurrent cell (GRU, RSSM or state-space) that **moves the state forward one real time step** (a day, an hour, six hours), taking in what is known about that step (weather forecast, calendar, ...). The state is **carried** from step to step: $z_{t+1} = f(z_t, u_t)$. This is the “imagination” of the model.
3. Readout	Turns each simulated state into numbers, usually through **physics or economics built into the layer** (mass balance, merit-order curve, stochastic volatility), so outputs cannot violate basic laws of the market.
4. Uncertainty	Every model outputs quantiles (a full range, e.g. 10%–90%), not one number; scored with the pinball/quantile score (QS).
5. Ensemble	Three independently trained seeds, averaged quantile-by-quantile; the audit follow-ups also blend with the best simple baseline (weights chosen on validation data only).
6. Direct twin	A control model with identical parts and size but **no carried state**; it tells us whether the world-model dynamics add anything (section 2.5, Figure A2).

2.3 Size and training of the frozen models (read from the checkpoint files)

Model	Market / target	Params per member	Members	Trained on data up to	Training
wm_gas	US gas storage, weekly	23,534	3 seeds	2025-06-30	1,200 steps, lr 3e-3, d=32
wm_gas_ens	same	23,534	14 + OLS	2025-06-30	pooled seeds 0–6 (rollout + direct)
wm_power	DE-LU power, hourly	91,845	3 seeds	2025-06-30	1,500 steps, lr 1e-3, d=64
wm_fin	crypto vol, daily	64,301	3 seeds	2025-06-30	800 steps, lr 2e-3, d=64, z=16
wm_sv	WTI oil, daily	25,191	3 seeds	2022-12-31	600 steps, lr 2e-3, 128-day input

These are tiny by modern standards (23K–92K parameters; Chronos-2 and TiRex have tens to hundreds of millions) and run on a laptop CPU in seconds.

2.4 Each model, explained

wm_gas — US natural-gas storage world model (and wm_gas_ens)

diagram: **A3, A4**

Predicts	The weekly change in US natural-gas storage (Lower 48 and 5 regions, in Bcf) that the EIA publishes every Thursday; 13 quantiles; forecasts made on Thursday and Friday before release.
Sees	Storage history released so far, the 5-year average for the week, observed daily heating/cooling degree days (NOAA CPC) up to the forecast day, the CPC 7-day weather forecast for the rest of the target week, normals, calendar and Henry Hub price history.
How the state works	An encoder makes the starting state. A GRU cell then steps it **one day at a time**: first through last week (with observed weather) — the model's own estimate of last week's storage change is compared with what EIA actually released, and the error is written back into the state (**data assimilation**, like a Kalman filter). Then it steps through the 7 days of the target week (“imagination”).
Built-in physics	Each day, each region's net injection = the 5-year-average flow + a learned base − demand terms that are **forced to rise with heating and cooling degree days**; 3 learned regimes; weekly change = sum of days; Lower 48 = sum of the 5 regions (**mass balance**).
Output	Median from the physics readout; intervals from held-out residual quantiles; 3 seeds averaged.
wm_gas_ens (audit)	14 members (rollout and direct, seeds 0–6) averaged, blended 0.8 / 0.2 with the OLS-with-weather baseline, and intervals scaled by the model's accuracy over the last 8 weeks; every choice fixed on development data before the test look.
October 2026	wm_gas_ens best quantile score of 13 models from both origins (Friday QS 2.75 vs OLS 3.87, Chronos-2 3.84).

wm_power — German day-ahead electricity world model (and wm_power_blend)

diagram: **A5**

Predicts	All 24 hourly prices (EUR/MWh) of tomorrow's German-Luxembourg day-ahead auction, 9 quantiles each, issued before the noon auction.
Sees	14 days of hourly prices up to yesterday, and grid fundamentals (load, wind, solar) only up to the day before yesterday (a deliberately conservative rule, because they are published late).
How the state works	Each hour of the 14-day history is embedded and read by a GRU into a state at the end of D−2; a GRU cell then steps it **hour by hour for 48 hours** (through D−1 into the delivery day), carrying the state.
Built-in physics	A **residual-load head** (demand minus wind and solar) and a **monotone merit-order curve**: the price can only rise as residual load rises, the way power plants are dispatched cheapest-first; plus 3 regime offsets.
Output	13 monotone quantiles per hour; 3 seeds averaged. wm_power_blend = 50/50 quantile blend with LEAR (the standard electricity-price benchmark), weight chosen on validation data.
October 2026	Lowest MAE (wm_power 32.79, blend 32.76 EUR/MWh); 4th on QS (13.10) behind Chronos-2 and Chronos-Bolt; far ahead of LEAR (15.24) and the naive baselines.

wm_fin — crypto volatility world model (and wm_fin_blend)	diagram: **A6**
Predicts	Tomorrow's realised volatility (from 5-minute returns) of 24 Binance perpetual futures, as 13 quantiles of log volatility, and whether it will be above its 7-day average.
Sees	14 days of daily features (volatility, returns, volume, buy/sell pressure, funding rate) and hourly features up to the 00:00 UTC decision time, plus the weekday.
How the state works	An **RSSM** (the design behind the Dreamer agents): a deterministic GRU memory plus a 16-dimensional random latent. It filters the 14 days (each day's data corrects the state), then **imagines tomorrow in four 6-hour steps** without seeing any data.
Output	Heads read the imagined state into 13 quantiles; 3 seeds averaged. Trained on 16 coins only — the other 8 coins were never seen in training, to test transfer. wm_fin_blend = 0.6 wm_fin + 0.4 har_intra_dow (a strong calendar-aware regression), weight chosen on validation data.
October 2026	The world-model family leads on quantile score (blend 0.1435, wm_fin 0.1490 vs har_intra_dow 0.1522, Chronos-2 0.1605); on direction lgbm_dow is ahead.

wm_sv — crude oil stochastic-volatility world model	diagram: **A7**
Predicts	The WTI spot price 1, 5 and 20 trading days ahead, as 13 quantiles.
Sees	Only the last 128 days of WTI prices.
How the state works	An observer GRU reads the history into a state holding a latent price level and a latent log-volatility. Each simulated day, a **return step** draws a fat-tailed (Student-t) return and a **volatility step** updates the log-volatility; the sampled return is fed back in (closed loop). Many sample paths are simulated.
Built-in finance	A stochastic-volatility model: calm and turbulent periods are represented explicitly, and the forecast range widens when volatility is high.
Output	Quantiles of the simulated paths; 3 seeds averaged in log space.
October 2026	1-day QS 0.00932 vs random walk 0.00933: a tie, as on the test year. Oil prices are close to a random walk at short horizons.

2.5 Does the world-model part earn its place? (the direct-twin test)

Each world model has a **direct twin**: same modules, parameter count, data, training steps and seeds, but the state is not carried from step to step — every step is one move from the starting state. Any gap between the two measures the value of the dynamics alone. October 2026:

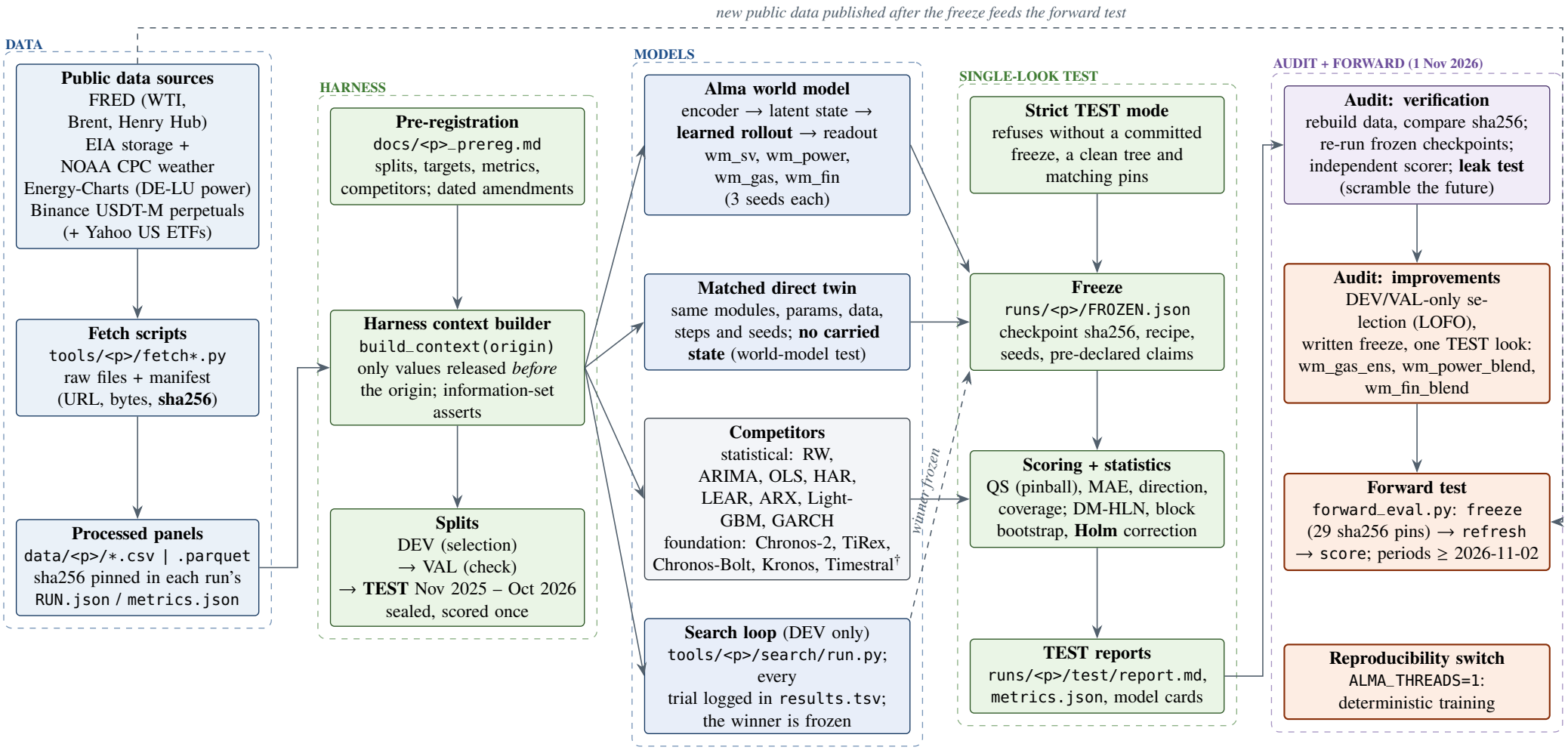
Metric (lower is better)	World model	Direct twin	Reading
Gas MAE thu / fri (Bcf)	8.19 / 8.08	8.63 / 8.72	rollout ahead, 5 weeks only
Power QS (EUR/MWh)	13.10	13.31	rollout ahead, 31 days
Crypto QS (log RV)	0.1490	0.1478	twin ahead
Oil 1-day QS (log price)	0.00932	0.00931	tie

All four gaps are inside noise. The world-model test (rollout beats its twin) is still **not passed**, consistent with the audited TEST year. What clearly does help is the rest of the design: the physics-shaped readouts, ensembling, and blending with a strong simple baseline.

The next 8 pages (Figures A1–A8) are the detailed architecture diagrams, drawn from the model source code (reports/Alma_Models_Architecture.pdf, commit 43b463b — the deployed checkpoints are the same frozen files, verified by sha256).

1. End-to-end system architecture

The pipeline every market project followed (oil, power, gas, crypto), with the audit and forward-test layers added on 1 November 2026. Source: `tools/<project>/`, `runs/<project>/`, `docs/*_prereg.md`, `tools/forward/`.



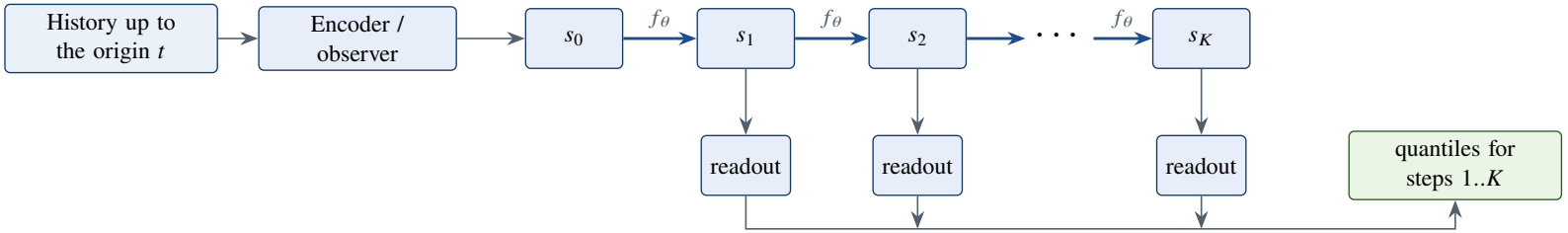
†Timestral-5M is scored but excluded from claims because its training data overlap the test period (flagged "contaminated" in every report).

data · original Alma model · added by the 1 Nov 2026 audit · evaluation · audit · baseline / other · latent rollout (carried state) · used in training only / optional

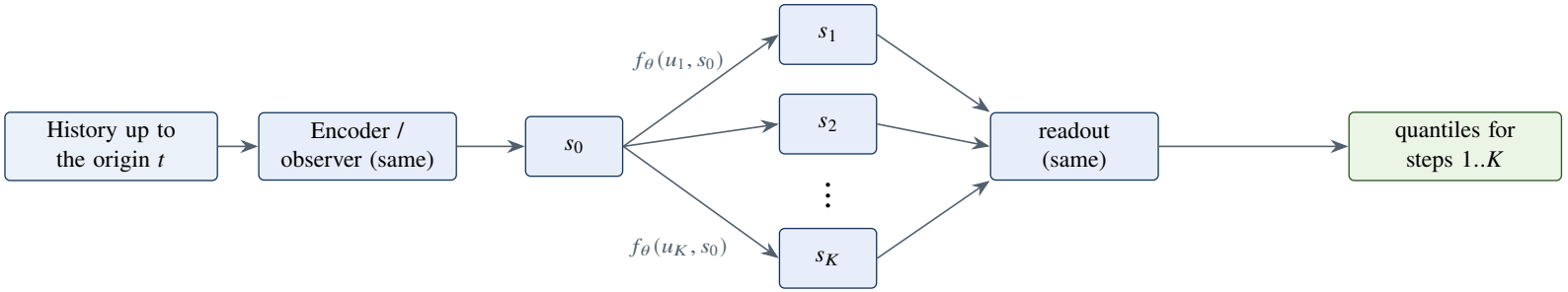
2. The core design: world model vs matched direct twin

Every Alma model has a twin with identical modules, parameter count, data, training steps and seeds. Only how the latent state advances differs. This comparison is the project's mandatory "world-model test" (`alma_goal.md` §2).

(a) World model (`arch=wm` / `abl=none`): $s_k = f_\theta(u_k, s_{k-1})$, the state is carried from step to step



(b) Direct twin (`arch=direct` / `abl=direct`): $s_k = f_\theta(u_k, s_0)$, every step is one move from the origin state



Pass rule (`alma_goal.md` §2)
The world model must beat its twin on the primary metric over ≥3 seeds, with a paired bootstrap CI that excludes 0.

Outcome on TEST
The twin is equal or better within noise in oil, power, gas and crypto.

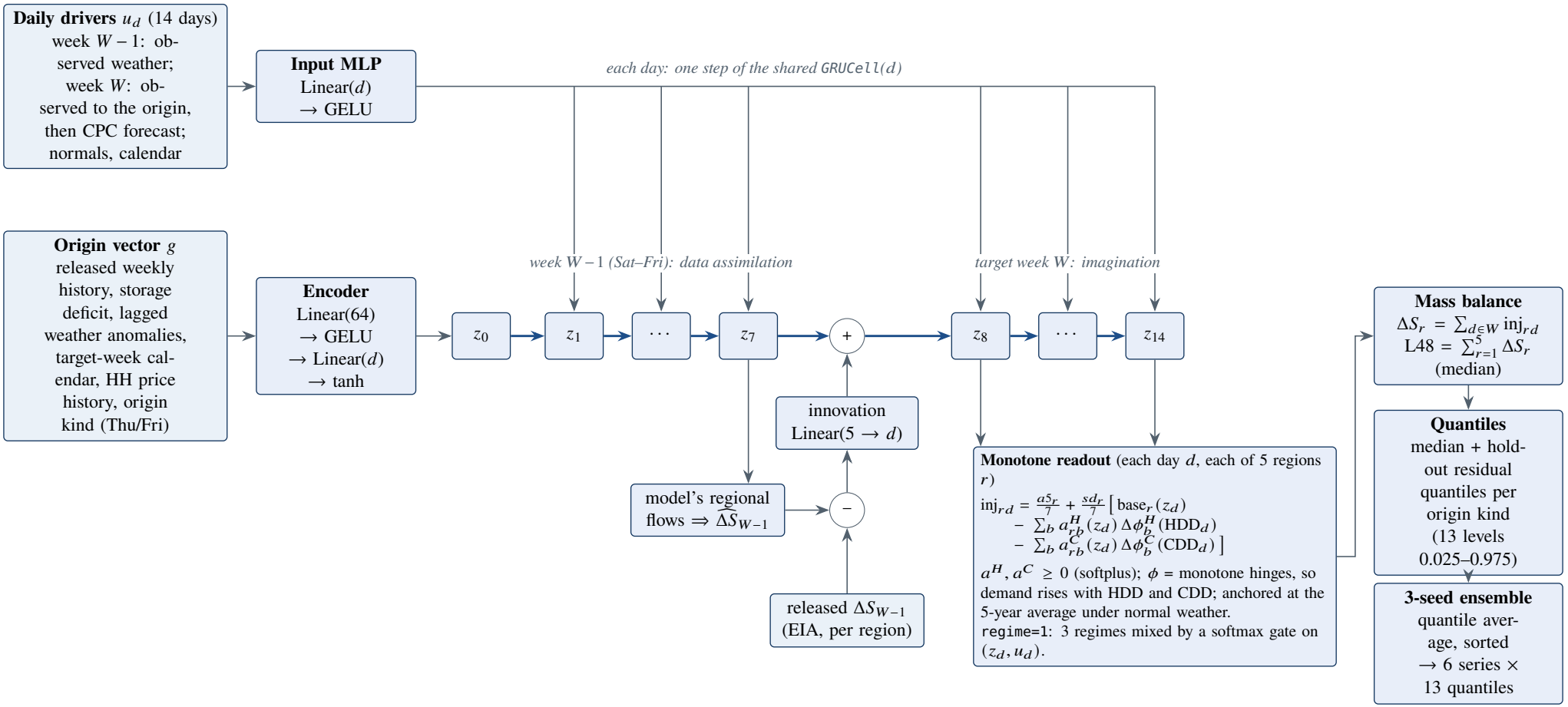
Research-track SSM, 5 seeds: rollout gain +0.7%, CI [−0.03%, +1.3%], not significant; +0.08% without the selection seed.

u_k is the step-k driver (calendar or phase clock, known or forecast exogenous inputs, lagged observations). Each project's twin is built by the same class with one flag changed: oil `SVWM(abl="direct")`, power `PhysWM(abl="direct")`, gas `GasNet(rollout=False)`, crypto `RSSM(abl="direct")`.

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3. wm_gas: storage-balance state-space world model (US natural-gas storage)

`tools/gas/search/nets.py` (`GasNet`), wrapper `tools/gas/wm_gas.py`. Frozen recipe: `arch=wm, assim=1, regime=1, d=32`, dropout 0.3, wd 0.05, 1,200 steps; seeds 0, 1, 2; trained on weeks released ≤ 2025-06-30.

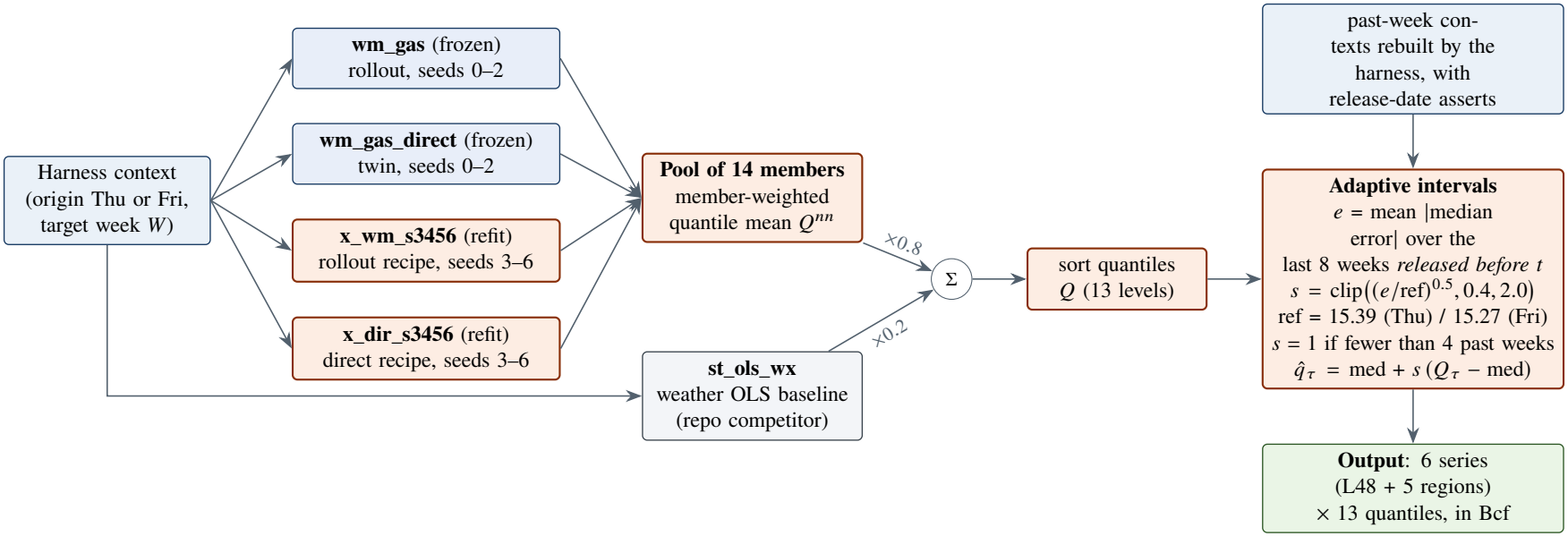


Twin (`wm_gas_direct`): each z_d is one GRUCell step from z_0 (no state carried between days). The innovation is computed in the same way, so both models see identical inputs. Size: about 23.5K parameters per member. **TEST** (Thursday, 52 weeks): MAE 11.53 Bcf, QS 4.59; the twin scores MAE 11.40, QS 4.57.

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4. wm_gas_ens: audit ensemble with OLS blend and adaptive intervals (new)

`tools/gas/wm_gas_ens.py`; checkpoints in `winning_models/gas_wm_gas_ens/` (sha256 asserted at load). Every choice was fixed on the DEV backtest (`audit/GAS_FREEZE.md`, written 16:34 and 17:00 UTC on 1 Nov 2026) before the TEST look.



How it was chosen (DEV, 6 rolling folds 2019-07 to 2025-06).

- The pool of rollout and direct seeds plus OLS, with the weight chosen leave-one-fold-out, reached DEV MAE 15.46 vs 16.23 for the frozen recipe with 7 seeds, lower in all 6 folds.
- Final frozen weight: 0.2 (full-DEV optimum); DEV MAE 15.39 / 15.27.
- Interval rule, chosen LOFO on DEV: DEV QS 5.970 → 5.919.

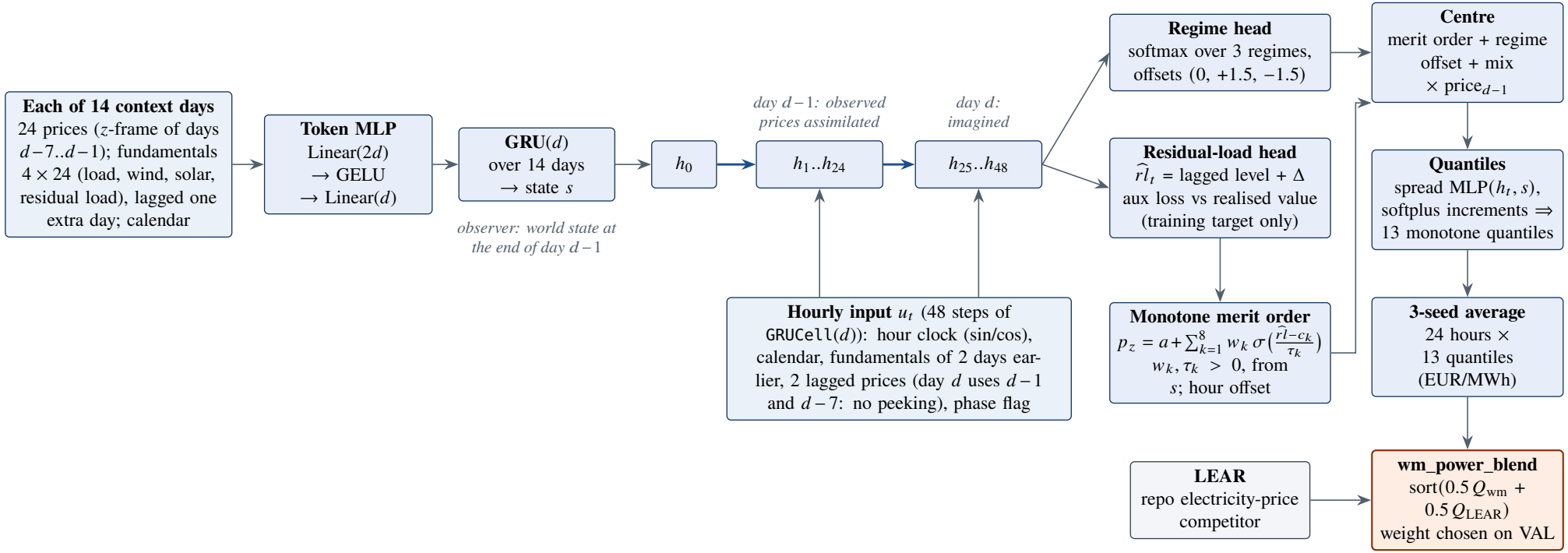
TEST result (52 weeks, Thursday / Friday).

- MAE 10.63 / 10.46 Bcf (wm_gas 11.53 / 11.56); QS 4.14 / 4.07 (4.59 / 4.56).
- Significant vs OLS on MAE and QS; not significant vs wm_gas.
- 80% coverage 0.885 / 0.904; intervals 17% narrower than with fixed widths.
- Leak test with the global data store scrambled: 0 of 22 forecasts changed.

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5. wm_power: physics-informed merit-order world model (German day-ahead power)

`tools/power/wm_models.py` (`PhysWM`, `arch=phys`), wrapper `tools/power/wm_power.py`. $d = 64$, 14 days of context, auxiliary loss weight 0.3, lr 10^{-3} OneCycle, batch 64, 1,500 steps, seeds 0–2. The blend was added by the audit.

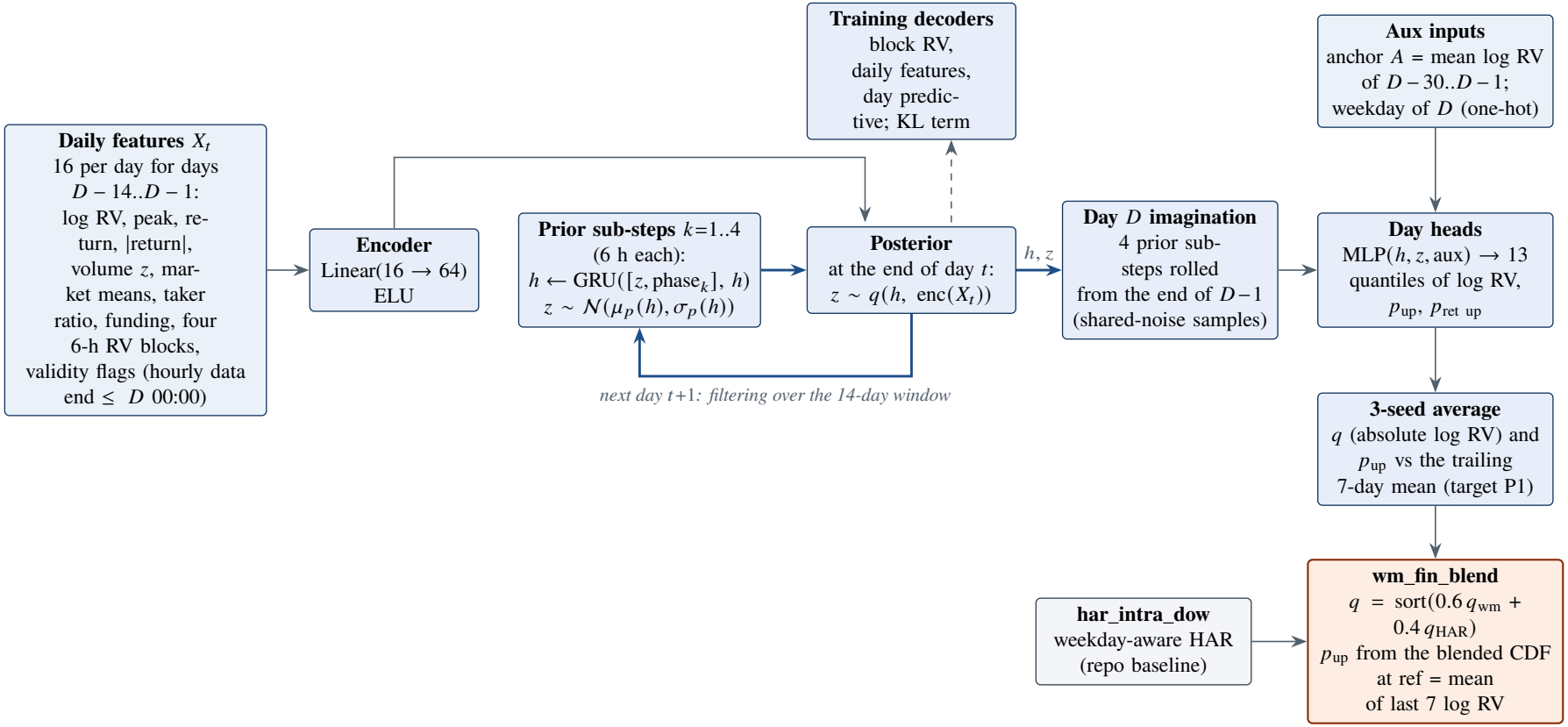


Twin (`wm_power_direct`): every hour is one GRUCell step from s. **TEST** (365 days): wm_power P2 0.747, QS 9.02, cov80 0.752; twin P2 0.739, QS 9.08. **Blend**: QS 8.60 (significant vs wm_power and LEAR), P2 0.755 (significant vs LEAR only).

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6. wm_fin: RSSM world model on an intraday clock (crypto realized volatility)

`tools/fin/wm_core.py` (`RSSM`, "DreamerV3-lite"), wrapper `tools/fin/wm_fin.py`. $d = 64$, $z = 16$, 4 prior sub-steps (6 h each) per day, window $W = 14$ days, weekday-of-D input; lr 2×10^{-3}, 800 steps; 3 seeds refit on labels $\leq$ 2025-06-30 (16 training coins). The blend was added by the audit.



Twin (`wm_direct`): the day heads read the filtered state at the end of $D-1$, with no day-D rollout (identical modules and losses).

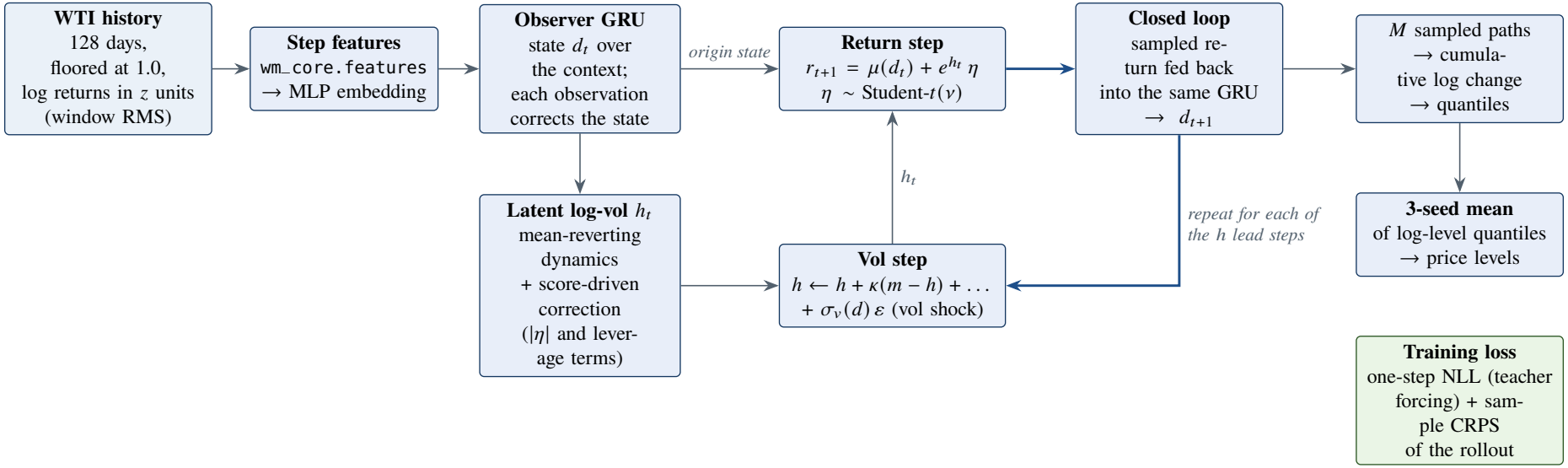
Other frozen pillars (not diagrammed): wm_fin_cl (continual SGD, 1 step per day), wm_agent (Dreamer-style actor-critic), PPO/DQN twins.

TEST (24 coins): wm_fin P1 0.727, QS 0.1471; twin P1 0.724, QS 0.1465; har_intra_dow P1 0.731. **Blend**: P1 0.740, QS 0.1459. Significant vs wm_fin on P1; not vs har_intra_dow.

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7. wm_sv: stochastic-volatility world model (WTI crude oil)

`tools/commodity/wm_models.py` (SVWM), wrapper `tools/commodity/wm_commodity.py`. Input: 128 days of WTI only. $d = 64$, about 25,191 parameters per member; lr 2×10^{-3} cosine, batch 64, 600 steps; training windows end before 2023-01-01; seeds 0–2, mean of log-level quantiles.

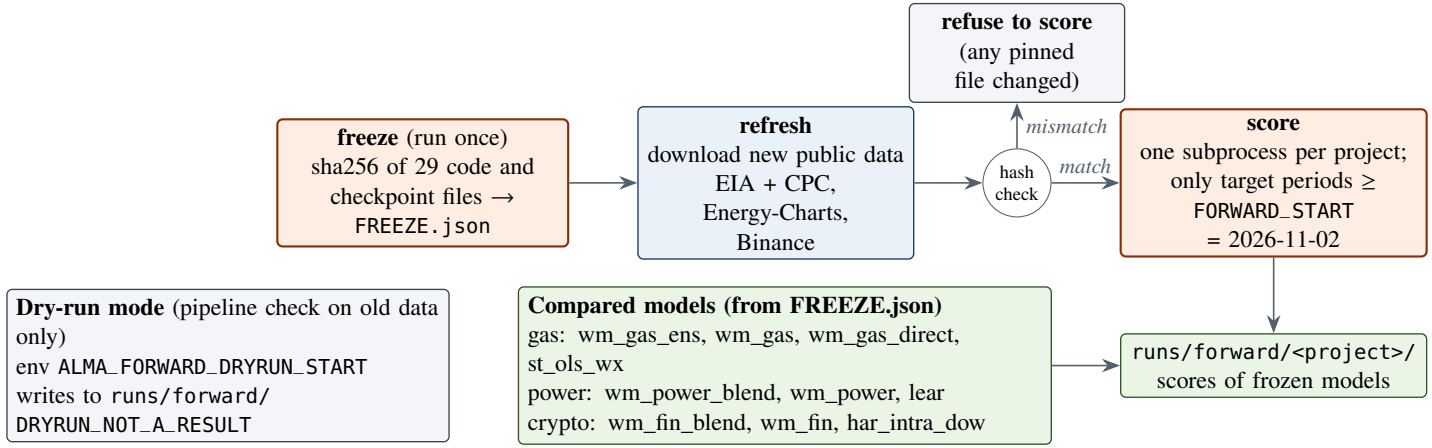


Twin (`wm_sv_direct`): lead k is one step from the origin state, $d_k = \mathrm{GRU}(\mathrm{emb}_k, d_{\mathrm{origin}})$, decoded as a Student-$t$ with scale $e^{h_{\mathrm{origin}} + m(d_k)}\sqrt{k}$; nothing is sampled step by step.
TEST ($h = 1$, 246 origins): QS 0.01052 vs random walk 0.01056 (skill +0.45%, $p = 0.479$, a tie); twin 0.01056. No improvement model was built, because the signal is indistinguishable from a random walk.

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8. Forward-test architecture (frozen 1 November 2026, 17:47 UTC)

`tools/forward/forward_eval.py`, `tools/forward/FREEZE.json`. Scores the frozen models only on data published after they were frozen.



Data arrival: first EIA storage report 2026-11-05; gas release #20 on 2027-03-18 and #52 on 2027-10-28; power and crypto day 60 on 2026-12-31 and day 365 on 2027-11-01. These follow the publication calendars, with holiday shifts possible.

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3. Deployment setup

	Dates
Training data ends	2025-06-30 (oil: 2022-12-31)
Validation / sealed TEST year	to 2025-10-31 / 2025-11-01 … 2026-10-31
Forward-test freeze	2026-11-01
“Last month” scored	October 2026 (last month of the TEST year, scored before)
“This month” scored	November 2026: oil to 2026-11-03, power to 2026-11-05, crypto to 2026-11-03; gas: no release yet

Competitors kept their own pre-registered protocols (for example, LEAR refits its lasso daily on data available at the origin; OLS/HAR were fitted on the training period by their own code). That is how those baselines are defined, not additional training of Alma.

4. Data rebuilt and verified

Project	Verification against the committed pins	Cutoff
Oil	FRED WTI/Brent/HH up to the pinned date: byte-identical	2026-11-03
Power	2015–2025 Energy-Charts files byte-identical; 2026 price file (pinned URL) byte-identical; extension matches every overlapping point	2026-11-05
Gas	weekly.csv and cpc_fcst.csv equal their pins; 4,784/4,786 CPC files identical (2 are growing 2026 files); 52-week TEST forecasts reproduced to 5.1e-05 Bcf	2026-10-29
Crypto	all 4,700 Binance files byte-identical; universe.json equal to its pin; October forecasts reproduce the audit's TEST cache	2026-11-03

5. Results against the competition

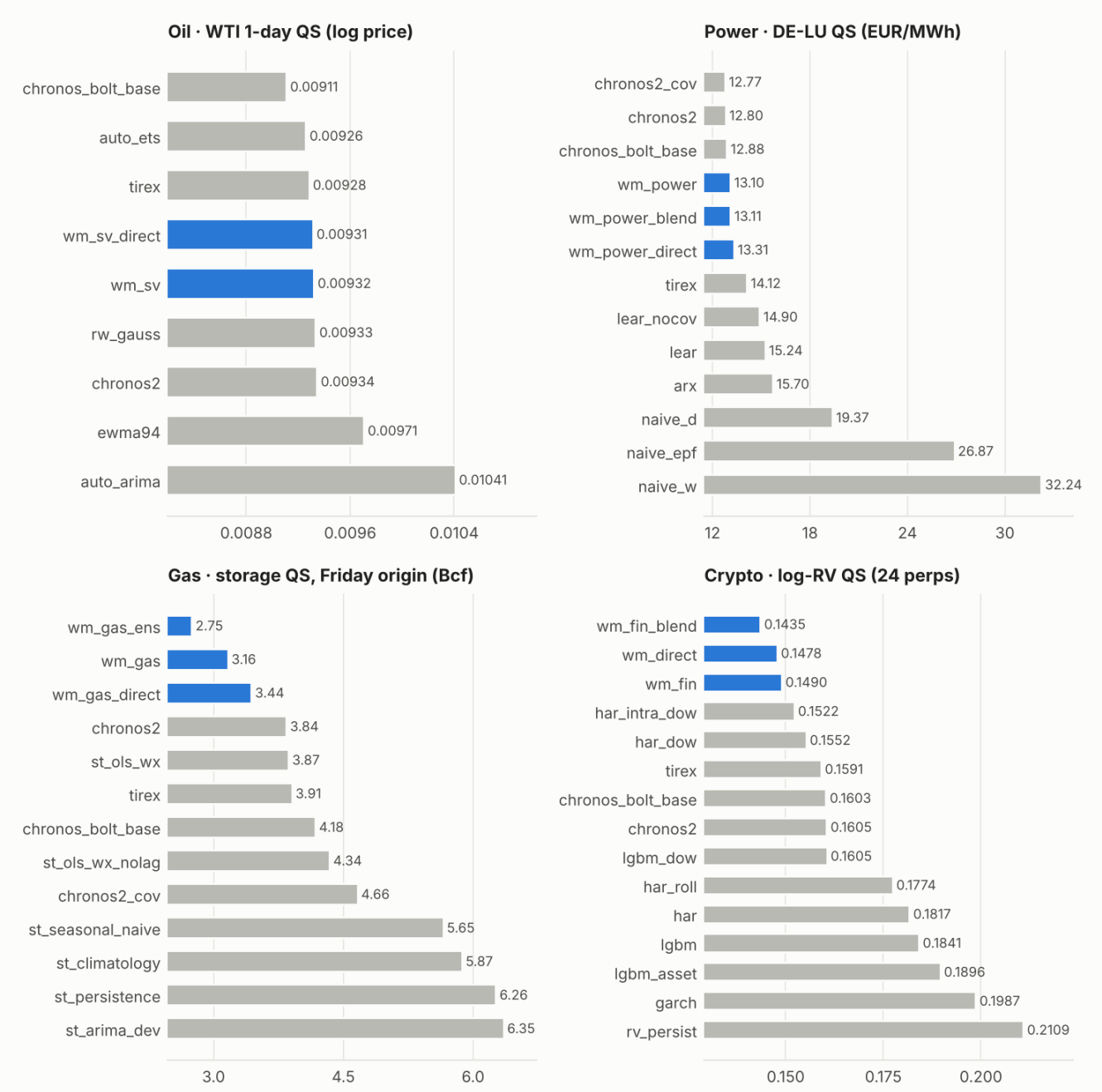


Figure 1. October 2026 quantile score (interval quality) by project.

Gas — EIA weekly Lower-48 storage change

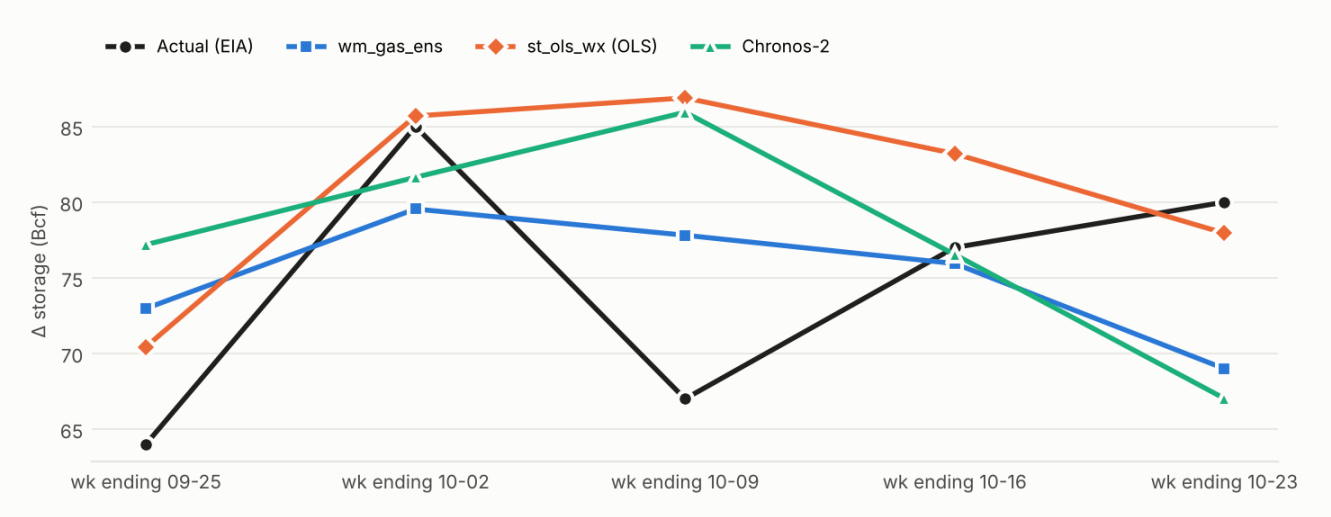


Figure 2. Thursday-origin median forecasts vs the EIA releases (2026-10-01, 2026-10-08, 2026-10-15, 2026-10-22, 2026-10-29).

Model	MAE thu	QS thu	MAE fri	QS fri
wm_gas_ens	7.46	2.76	7.24	2.75
wm_gas	8.19	3.15	8.08	3.16
wm_gas_direct	8.63	3.48	8.72	3.44
chronos2	9.78	3.84	9.78	3.84
st_ols_wx	7.05	3.85	7.75	3.87
tirex	10.16	3.91	10.16	3.91
chronos_bolt_base	10.55	4.18	10.55	4.18

Top 7 of 13 by Thursday QS (full table in RESULTS.md). Highlighted rows are Alma models. wm_gas_ens has the best QS from both origins; on MAE it is 2nd behind OLS on Thursday and 1st on Friday — the same pattern as the TEST year.

Sealed ex-ante forecast — week ending 2026-10-30, EIA release 2026-11-05 (5-year average 36.4 Bcf)

Model	Thursday origin: median [80%]	Friday origin: median [80%]
wm_gas_ens	74.2 [56.9, 91.5]	74.9 [57.8, 92.0]
wm_gas	74.9 [52.3, 97.5]	75.8 [53.2, 98.1]
st_ols_wx	75.6 [46.0, 107.3]	75.5 [45.4, 107.1]
chronos2	53.0 [17.4, 80.5]	53.0 [17.4, 80.5]
tirex	57.6 [19.8, 86.8]	57.6 [19.8, 86.8]

Crypto volatility — 24 Binance perpetuals

Model	QS (P3)	Direction acc. (P1)	P1 held-out coins	80% coverage
wm_fin_blend	0.1435	0.694	0.745	0.86
wm_direct	0.1478	0.711	0.745	0.82
wm_fin	0.1490	0.705	0.750	0.82
har_intra_dow	0.1522	0.674	0.685	0.89
har_dow	0.1552	0.696	0.708	0.89
tirex	0.1591	0.673	0.694	0.82
chronos_bolt_base	0.1603	0.684	0.713	0.81
chronos2	0.1605	0.681	0.713	0.78

October, corrected data, top 8 of 15. The world-model family leads on QS; lgbm_dow leads on direction. November (3 days): wm_fin has the best QS (0.1104); no direction labels exist yet because of the 28 Oct archive gap.

Power — German day-ahead, 24 hourly prices

Model	QS (EUR/MWh)	MAE	Hourly dir. (P2)	Daily dir. (P1)
chronos2_cov	12.77	32.99	0.762	0.677
chronos2	12.80	33.17	0.766	0.677
chronos_bolt_base	12.88	32.83	0.792	0.742
wm_power	13.10	32.79	0.722	0.710
wm_power_blend	13.11	32.76	0.750	0.710
wm_power_direct	13.31	33.33	0.719	0.710
tirex	14.12	36.63	0.749	0.677
lear_nocov	14.90	37.06	0.720	0.742

October, top 8 of 13. wm_power and its blend have the lowest MAE but rank 4th–5th on QS behind three foundation models; all models are far ahead of the naive baselines. November (5 days): Chronos-2 leads (QS 5.59 vs wm_power 9.42).

Oil — WTI spot

Model	1-day QS	skill vs RW	5-day QS	20-day QS
chronos_bolt_base	0.00911	0.024	0.02058	0.04204
auto_ets	0.00926	0.008	0.01856	0.02917
tirex	0.00928	0.005	0.02027	0.03304
wm_sv_direct	0.00931	0.002	0.01807	0.02699
wm_sv	0.00932	0.001	0.01851	0.02644
rw_gauss	0.00933	0.000	0.01913	0.03046
chronos2	0.00934	-0.001	0.01831	0.02934
ewma94	0.00971	-0.040	0.02041	0.03047
auto_arima	0.01041	-0.115	0.01824	0.02843

October, 22 targets per horizon. At 1 day wm_sv ties the random walk, exactly as on the TEST year. At 20 days it is numerically best, but 22 overlapping 20-day targets are about one independent observation, so that is not evidence. A sealed ex-ante WTI forecast is stored for the next prints.

6. The integrity harness and audit

Two layers. **Guards inside every run** (tools/deploy/common.py) make training impossible, hash all parameters, verify checkpoint pins and refuse significance verdicts on fewer than 10 independent units. **An independent audit** (tools/deploy/audit_deploy.py) shares no code with the deploy scripts, reads the saved forecasts and the RAW downloads, and fails loudly on any mismatch.

	Check group	Passed	Examples
A	Pins and frozen code	4/4	all winning_models checkpoints match SHA256SUMS and FREEZE.json (18 files); all FREEZE.json files (code + checkpoints) unchanged (29 files, changed: [])
B	No retraining	8/8	deploy scripts contain no optimizer/backward/train() call ([]); oil/results.json: torch parameters unchanged after deployment (7 modules hashed)
C	Truth from raw files and reproduction	15/15	oil: origin dates re-derived from raw FRED csv; oil oct_2026 h1: n targets from raw csv (n=22)
D	Every metric recomputed	5/5	oil: every QS recomputed from saved forecasts (54/54 match); power: every QS recomputed from saved forecasts (52/52 match)
E	Leak tests (future scrambled)	4/4	oil: future-scramble leak test (wm_sv, wm_sv_direct) (0/30 forecasts changed); power: future-scramble leak test (prices >= D, fundamentals >= D-1) (0/12 forecasts changed)
F	Sample-size rule	5/5	oil: every significance verdict rests on >= 10 independent units (48 verdicts, 0 significant, bad=[]); power: every significance verdict rests on >= 10 independent units (96 verdicts, 0 significant, bad=[])
G	Sealing	6/6	oil preds sha256 matches results; power preds sha256 matches results
R	Saved forecast = fresh model re-run	10/10	oil: saved wm_sv forecasts = fresh re-run of the frozen model (10 origins, max |diff| 0.0e+00); oil: saved wm_sv_direct forecasts = fresh re-run of the frozen model (10 origins, max |diff| 0.0e+00)
	Total	**57/57**	

Plus: tools/deploy/check_doc_numbers.py traces every decimal number in docs/deploy_2026_10_11.md to the outputs; a deliberately planted fake number was caught.

7. Findings raised by the audit

None is a leak or a fabricated number. Each was traced to a root cause:

#	Finding	Impact	Action
1	October 2026 crypto funding rates were missing from the pinned TEST data (published after the freeze) and silently set to 0; not listed in the manifest's missing list	Negligible: wm_fin QS 0.1430 → 0.1431 on identical days; funding-free models bit-identical	Both variants scored and reported
2	Binance's archive has no data at all for 2026-10-28 (0 bars, all 24 coins). forward_eval.py uses a row-based label reference that silently skips it; the official label leaves those days unlabelled	Small: 7 decision days scored differently by the forward test	Official labels used here; fix suggested as a separate task
3	Parquet bytes differ across platforms (arm64 vs x86-64)	None: forecasts reproduce the audit's cache	Content verified via reproduction
4	The power hourly.csv amendment pin cannot be reproduced byte-for-byte from today's downloads	None: raw files and values identical	Documented
5	Open-Meteo rate-limits this environment (HTTP 429), so gas daily.csv has no weather columns	None for the storage models (they don't read them); proven by exact reproduction	Documented; the HH-price task was not deployed

7b. Sealed forecasts graded so far

Written and sha256-sealed before the outcome existed; truth re-derived from the raw EIA / Energy-Charts files. Each row is one week or one day (n = 1): an illustration, not evidence.

Target	Actual	Alma model: sealed median	Alma rank (QS)	Best model (QS)
gas: week 2026-10-30 (thu origin)	77.00	wm_gas_ens 74.21	1 of 13	wm_gas_ens (1.842)
gas: week 2026-10-30 (fri origin)	77.00	wm_gas_ens 74.88	1 of 13	wm_gas_ens (1.750)
power: delivery 2026-11-06 (24 h)	275.64	wm_power 265.65	5 of 10	chronos2_cov (4.283)

108 model-forecasts graded across 3 targets; 6 sealed targets still await publication. Full table: runs/deploy_daily/EXANTE_LEDGER.md.

8. Progress accomplished

- The frozen models are now shown to **run unchanged on data 15–16 months after their training ended** (almost four years for oil) and to stay in the competitive pack on every project.
- The **data pipeline is proven reproducible** from public sources in a fresh environment, byte-for-byte where the format allows and forecast-for-forecast where it doesn't.
- There is now a **reusable, automated deploy → audit → report chain** (tools/deploy/run_all.sh) that refuses to train, refuses to over-claim and fails on any integrity breach.
- **Five data-quality issues** that the earlier audit did not record are now documented, with their measured impact.
- **Sealed forecasts** exist for targets not yet published, so the first truly out-of-sample grades cannot be tuned after the fact.

9. Next steps

- **2026-11-05**: EIA publishes the week ending 2026-10-30. Run tools/deploy/score_exante.py (refuses if a sealed file changed) and tools/forward/forward_eval.py.
- Keep the forward test running: about 20 gas weeks or 60 power/crypto days are needed before any confidence interval is informative.
- Fix the forward-test label reference (finding 2) before November crypto days are scored.
- Review and merge PR #2.